EDA_Optimising_NYC_Taxis_Rishabh_Kataria

Exploratory Data Analysis

Optimising NYC Taxi Operations

Rishabh Kataria

Dataset	NYC Yellow Taxi Trip Records — 2023 (12 monthly Parquet files)
Sample Size	265,487 trips sampled (≈0.7% per hourly stratum); cleaned to 256,490
Scope	Temporal, financial, geographical, operational & pricing analysis
Tools	Python · Pandas · NumPy · Matplotlib · Seaborn · GeoPandas

1. Data Preparation

The 2023 NYC yellow taxi dataset comprises 12 monthly Parquet files. Each individual month's file contains over 3 million records, making it computationally prohibitive to process all twelve months in their entirety. A stratified sampling strategy was therefore adopted to create a representative combined dataset.

Sampling and Combining Files

For each monthly Parquet file, trips were grouped by pickup date and pickup hour. From each date-hour group, 0.7% of trips were sampled at random (random_state=42). This hourly stratification ensures that the sample faithfully preserves the within-day temporal distribution of demand across all twelve months. The sampled portions from each month were concatenated into a single DataFrame, which was then serialised to a Parquet file for subsequent reuse.

The combined dataset contained **265,487 trip records** spanning all 12 months of 2023. Each original month contributed approximately 21,000–23,000 rows, ensuring roughly equal month-level representation.

File	Cumulative Rows After Sampling
2023-1.parquet	21,288
2023-2.parquet	21,491 (cumulative: 115,780)
2023-6.parquet + …	All 12 months combined: 265,487

Assumption: A 0.7% hourly-stratified sample is assumed to be statistically representative of overall 2023 trip patterns. Scale-up factor of ~143× (1/0.007) is applied where absolute volume estimates are needed.

2. Data Cleaning

2.1 Fixing Columns

2.1.1 Index Fix: The original DataFrame had a non-sequential index inherited from the source Parquet files. The index was reset to a clean RangeIndex (0 to 265,486) using reset_index(drop=True).

2.1.2 Duplicate Airport Fee Columns: The dataset contained two airport-fee columns — airport_fee (all-lowercase) and Airport_fee (title case) — likely due to inconsistency between vendor data feeds. The two columns were reconciled: where airport_fee was null, its value was filled from Airport_fee; the Airport_fee column was then dropped. The merged column retained the name airport_fee.

2.1.3 Negative Monetary Values: A sweep of all monetary columns revealed a small number of negative entries — the result of data entry or metering errors. These were clipped to zero rather than dropped, since the trip records themselves were otherwise valid.

Column	Negative Values Found	Action
fare_amount	0	No change
extra	1	Clipped to 0
mta_tax	11	Clipped to 0
tip_amount	0	No change
tolls_amount	0	No change
improvement_surcharge	11	Clipped to 0
total_amount	11	Clipped to 0
congestion_surcharge	6	Clipped to 0
airport_fee	2	Clipped to 0

RatecodeID contained 1,530 records with value 99, which is undefined per the TLC data dictionary. These were treated as missing during imputation.

2.2 Handling Missing Values

Five columns exhibited missing values, all with the same proportion of 3.33% (8,831 rows). This clustering strongly suggests that a specific vendor or dispatch system failed to transmit supplementary fields for this subset of trips.

Column	Missing (%)	Imputation Strategy
passenger_count	3.33%	Mode imputation (mode = 1); zero values also replaced with mode
RatecodeID	3.33%	Mode imputation (mode = 1.0 — Standard Rate); 99-coded records treated as missing
store_and_fwd_flag	3.33%	Dropped — not used in subsequent analysis

congestion_surcharge	3.33%	Filled with 0 — absence implies no surcharge was applied
airport_fee	3.33%	Part of combined airport_fee column; 0-fill applied

Zero passenger_count values (4,070 rows) were also replaced with the mode (1). After all imputation, 0 null values remained in the dataset.

2.3 Handling Outliers

Descriptive statistics and domain knowledge were used to identify implausible records. The following rules were applied:

Outlier Condition	Action	Rationale
trip_distance < 0.1 mi AND fare_amount > $300	Drop	Impossible for a short trip to cost >$300 legitimately
trip_distance = 0 AND fare_amount = 0 AND PU ≠ DO zone	Drop	Zero-fare, zero-distance across zones indicates a ghost record
trip_distance > 250 miles	Drop	Exceeds any realistic NYC taxi range (~2,600 rows removed)
payment_type = 0	Drop	Undefined in data dictionary
passenger_count > 6	Drop	Exceeds TLC-licensed sedan capacity (max 5-6 passengers)
trip_duration > 1440 minutes	Drop	Single trip lasting over 24 hours is operationally implausible

After applying all cleaning and outlier rules, the final clean dataset contained **256,490 records** — a retention rate of ~96.6%.

3. Exploratory Data Analysis

3.1 General EDA: Finding Patterns and Trends

3.1.1 Variable Classification

Type	Variables
Categorical	VendorID, RatecodeID, PULocationID, DOLocationID, payment_type, store_and_fwd_flag
Numerical (Continuous)	trip_distance, fare_amount, tip_amount, tolls_amount, total_amount, extra, mta_tax, improvement_surcharge, congestion_surcharge, airport_fee, trip_duration
Numerical (Discrete)	passenger_count, pickup_hour
Datetime	tpep_pickup_datetime, tpep_dropoff_datetime

3.1.2 Temporal Analysis — Pickups by Hour, Day and Month

Pickup demand was analysed across three temporal dimensions to understand how taxi usage varies within a day, across the week, and over the calendar year.

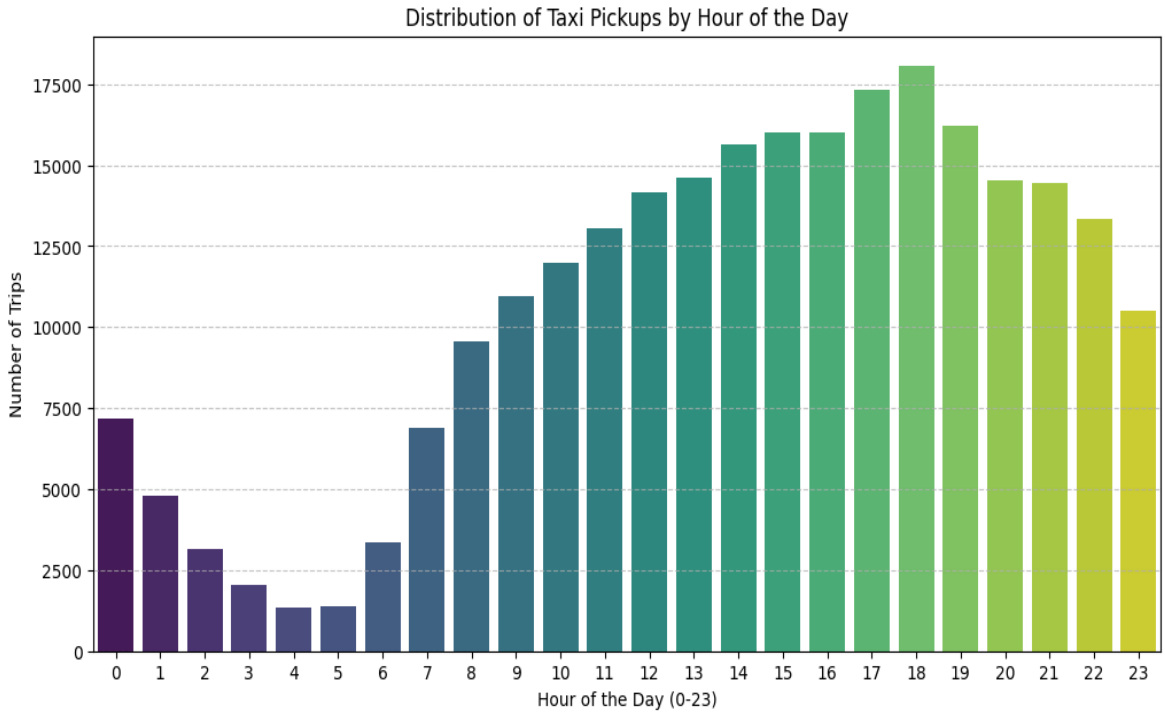

Figure 1: Hourly distribution of taxi pickups — demand peaks sharply at 18:00 (6 PM)

The busiest hour is **18:00** with 18,074 sampled trips, corresponding to the end-of-business rush. A secondary morning peak appears at 08:00–09:00. Overnight hours (02:00–05:00) record the lowest demand.

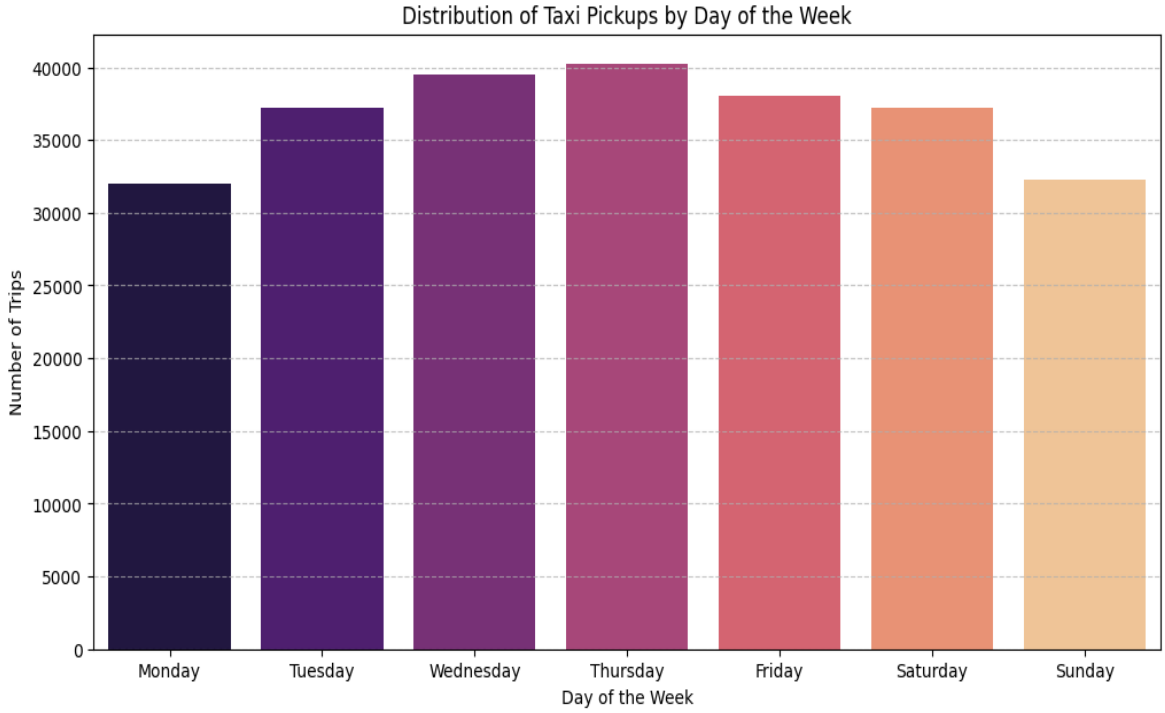

Figure 2: Pickups by day of week — Thursday leads, weekends slightly lower than weekday peak

Thursday is the busiest weekday (40,260 sampled trips), while Sunday records the fewest rides. Mid-week (Tuesday–Thursday) consistently outperforms the rest of the week.

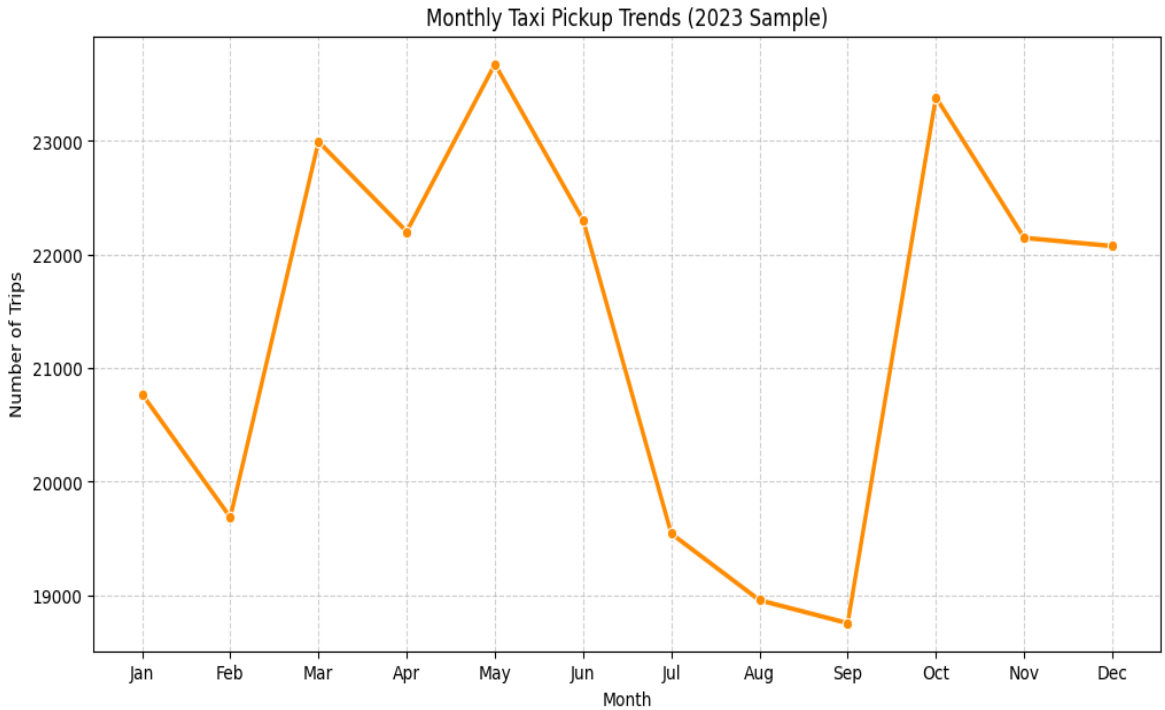

Figure 3: Monthly pickup distribution — May is the busiest month; summer months are strong

May is the busiest month (23,676 sampled trips). The warmer months of April–October sustain higher demand. February records the fewest trips, attributable to the shortest calendar month and winter weather.

3.1.3 Filtering Zero/Negative Values

Financial parameters were examined for zero and negative values before revenue analysis:

Column	Zero/Negative Count	% of Total
fare_amount	45	0.02%
tip_amount	57,265	22.33% — expected (cash payers do not tip via meter)
total_amount	0	0.00%
trip_distance	3,056	1.19% — airport queue/staging trips

A filtered DataFrame (df_non_zero) was created excluding records where fare_amount = 0 or trip_distance = 0, retaining 253,399 records for rate and financial analysis. This removed 3,091 records (1.2%).

3.1.4 Monthly Revenue Trends

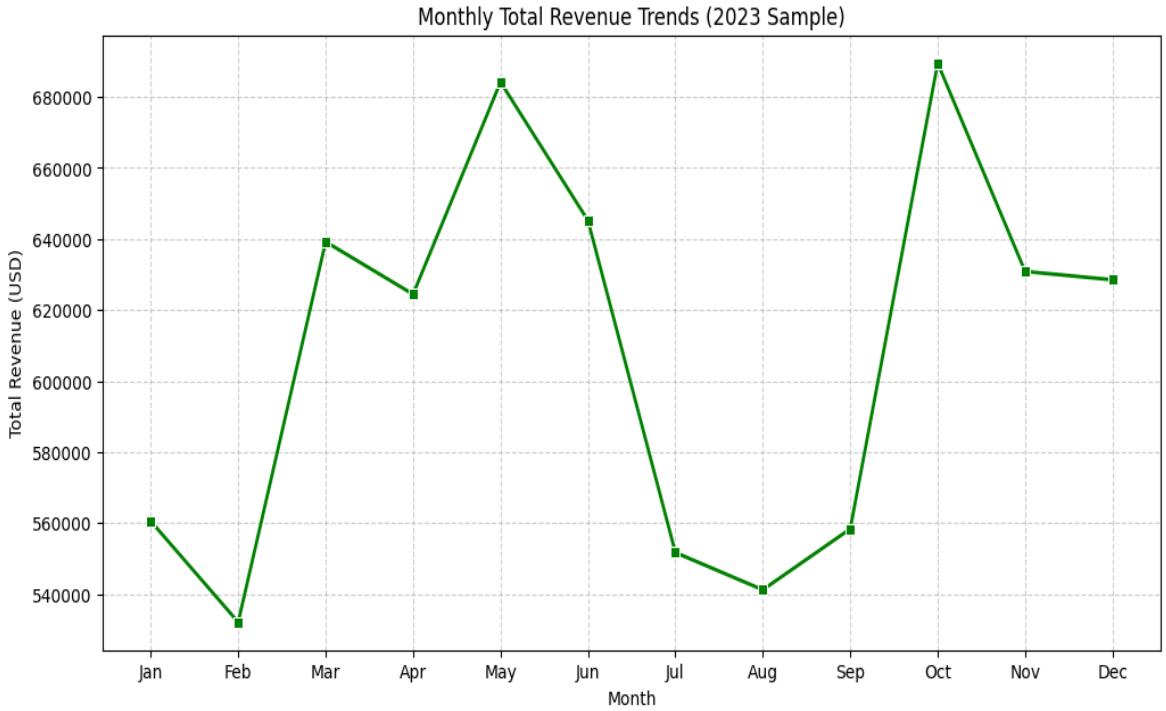

Figure 4: Monthly total revenue (sampled) — October is the highest-revenue month

Revenue (total_amount) tracks pickup volume closely but is additionally driven by average trip length. **October** generated the highest revenue at $689,504 in the sample, while February was the weakest. The strong Q2 and Q4 performance (see below) reflects both volume and fare levels.

3.1.5 Quarterly Revenue Share

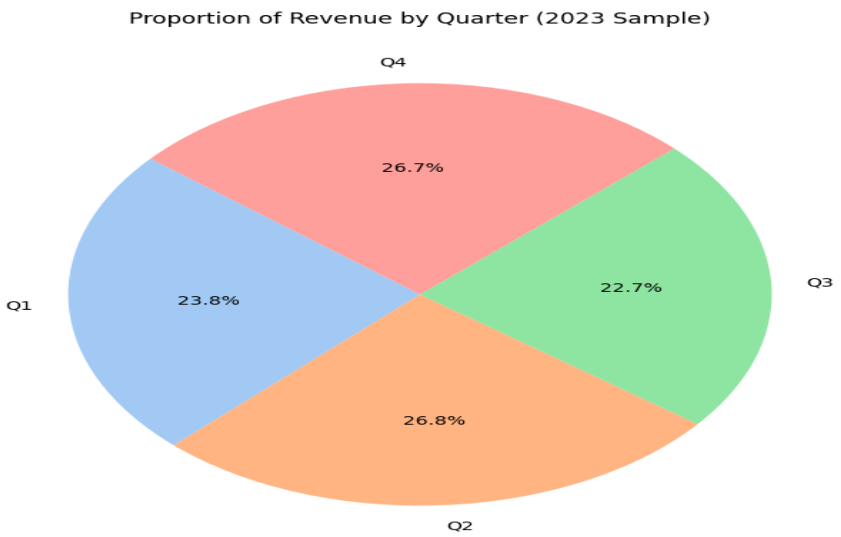

Figure 5: Quarterly revenue as a proportion of annual revenue

Quarter	Revenue (Sample, $)	Share (%)
Q1 (Jan–Mar)	1,731,925	24.2%
Q2 (Apr–Jun)	1,953,846	27.3%
Q3 (Jul–Sep)	1,651,376	23.1%
Q4 (Oct–Dec)	1,948,896	27.2%

Q2 and Q4 together account for ~54.5% of annual revenue. Q3 underperforms despite summer tourism, possibly due to residents vacationing and a shift towards ride-hailing alternatives.

3.1.6 Distance vs Fare Relationship

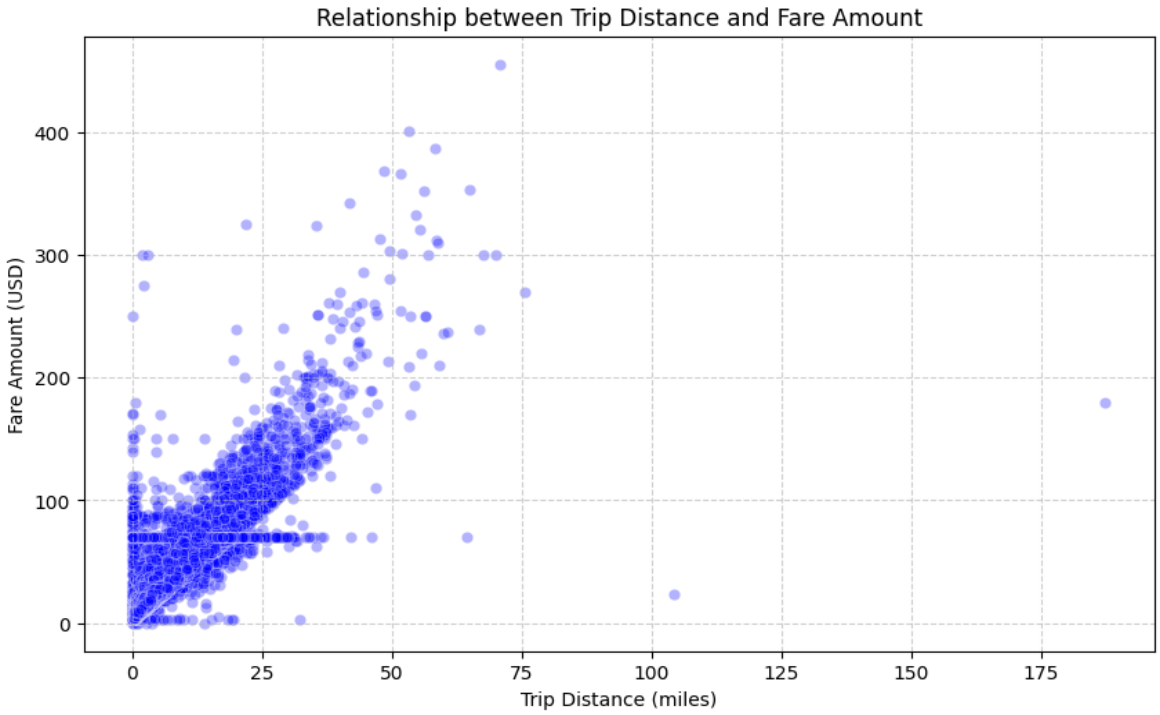

Figure 6: Scatter plot of trip distance vs fare amount — strong positive linear correlation

The Pearson correlation between trip_distance and fare_amount is **r = 0.943**, indicating a very strong positive relationship. This confirms that distance is the primary driver of fare. Some vertical scatter at low distances reflects flat-rate charges and minimum fare floors.

3.1.7 Fare/Tips vs Duration and Passengers

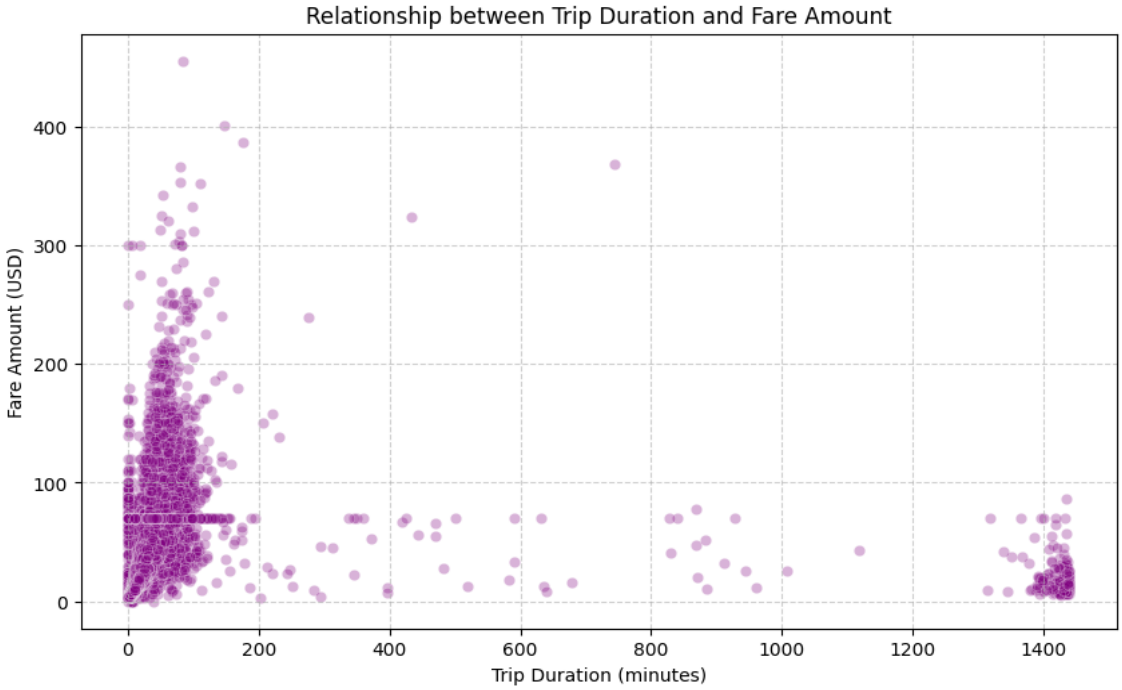

Figure 7: Trip duration vs fare amount — moderate positive correlation (r = 0.287)

Trip duration has a moderate positive correlation with fare (r = 0.287). The relatively weaker relationship compared to distance reflects stop-and-go urban traffic where meters tick by both distance and time, but the time component adds noise.

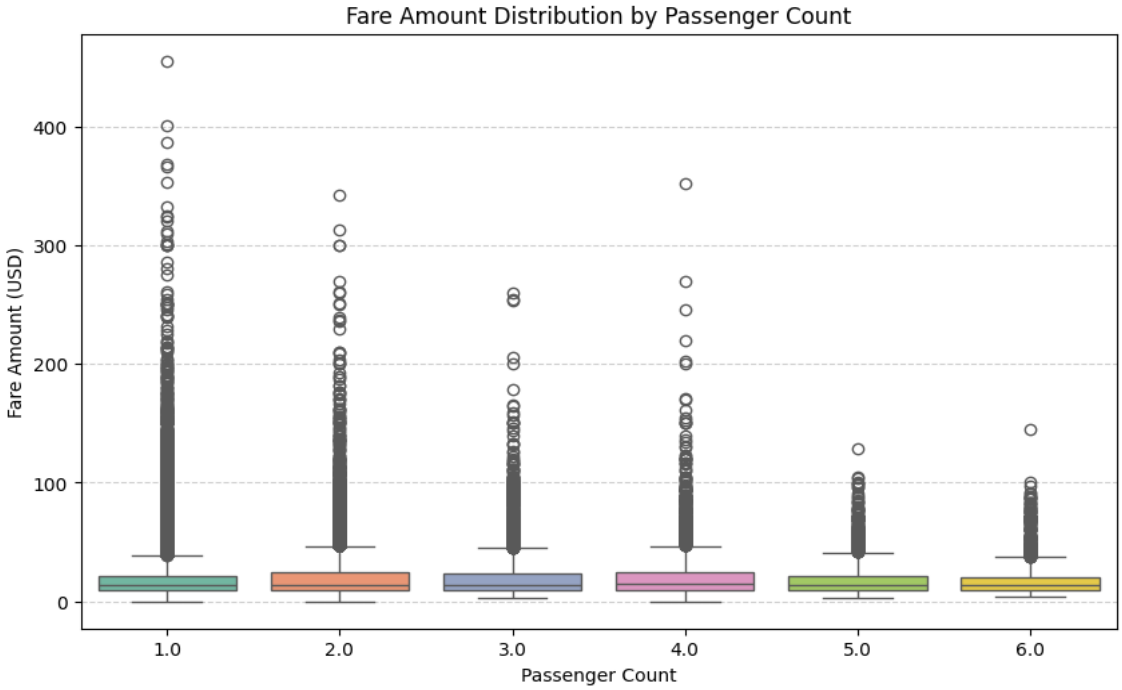

Figure 8: Fare by passenger count — fare is largely invariant to group size

The correlation between passenger_count and fare_amount is essentially zero (r = 0.042). NYC taxi fares are metered by distance/time, not headcount, so group size does not drive fare variation.

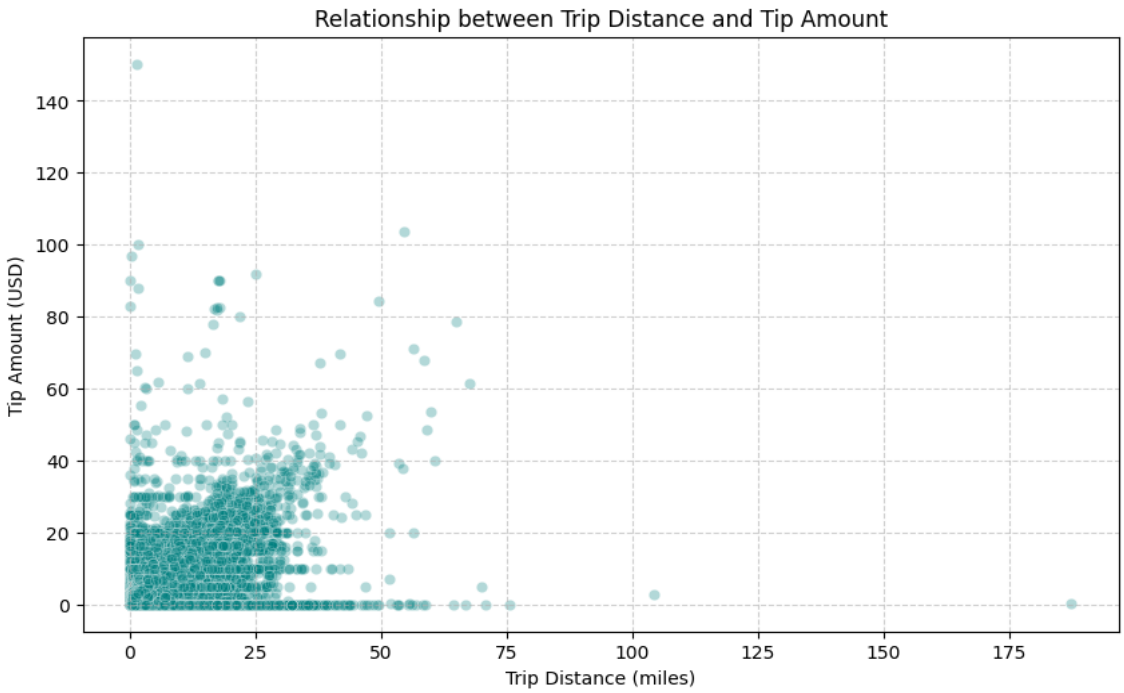

Figure 9: Trip distance vs tip amount — positive relationship (r = 0.590)

Trip distance and tip amount are moderately correlated (r = 0.590). Longer trips tend to produce higher absolute tip amounts, though the tip percentage actually decreases with distance (see Section 3.2.13).

3.1.8 Payment Type Distribution

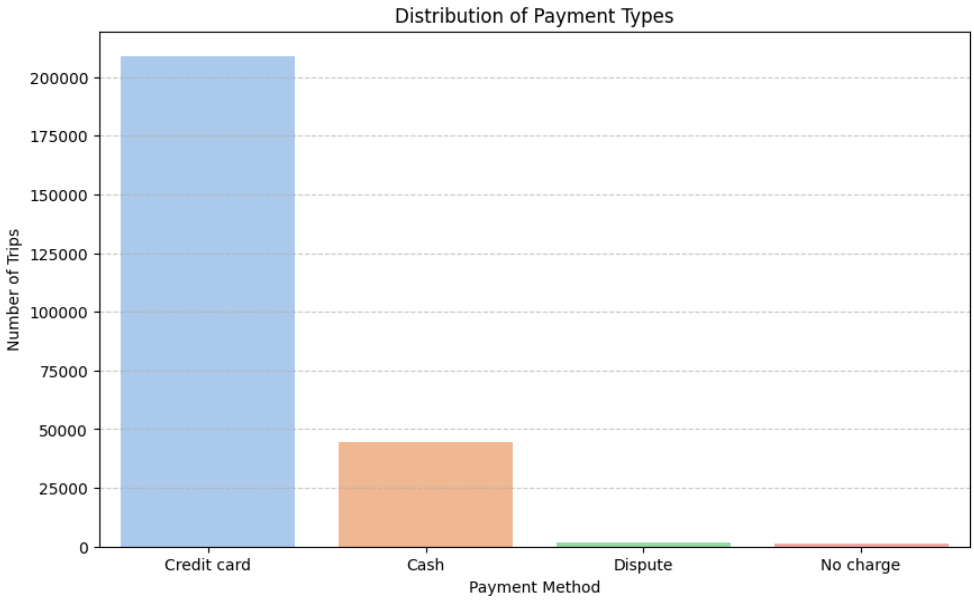

Figure 10: Distribution of payment types — credit card dominates at 81.5%

Payment Type	Share (%)
Credit Card	81.49%
Cash	17.29%

Dispute	0.74%
No Charge	0.47%

Credit card is the overwhelmingly dominant payment method (81.5%). This has implications for revenue analysis since tip data is only reliably captured for card transactions — the 17.3% of cash trips appear as zero-tip in the dataset, explaining the high proportion of zero-tip records noted earlier.

3.1.9–3.1.13 Geographical Analysis

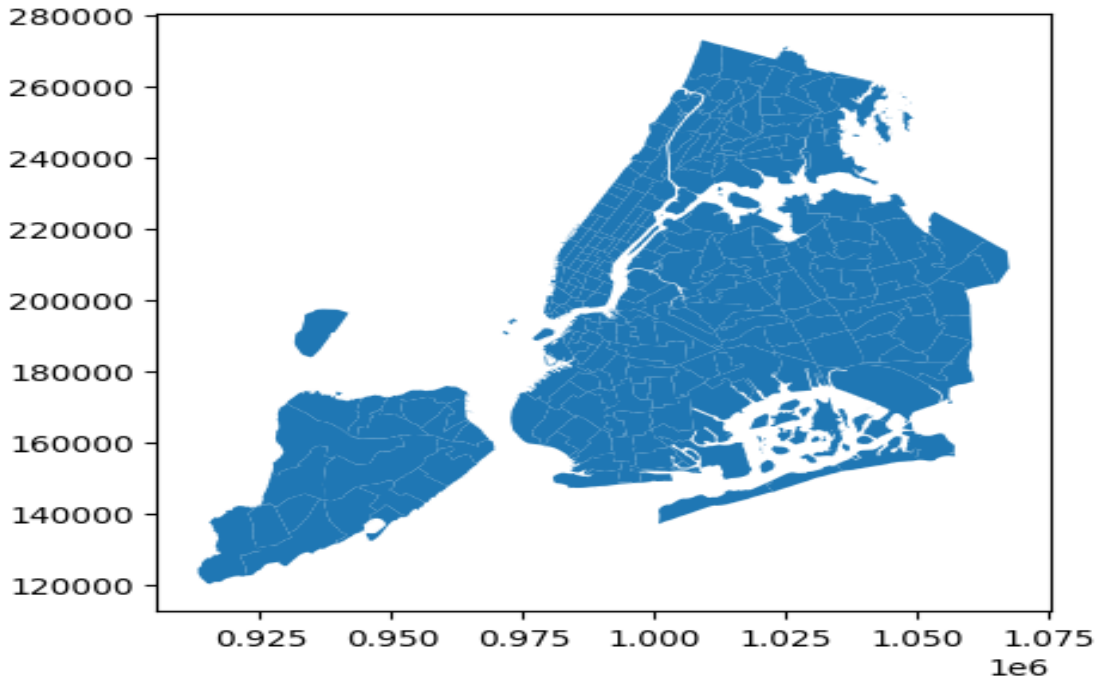

Figure 11: NYC taxi zone shapefile — 263 defined zones across all five boroughs

The taxi zones GeoDataFrame contains 263 zones with associated LocationIDs, borough labels and polygon geometries. Trip records were merged with zone data using PULocationID = LocationID.

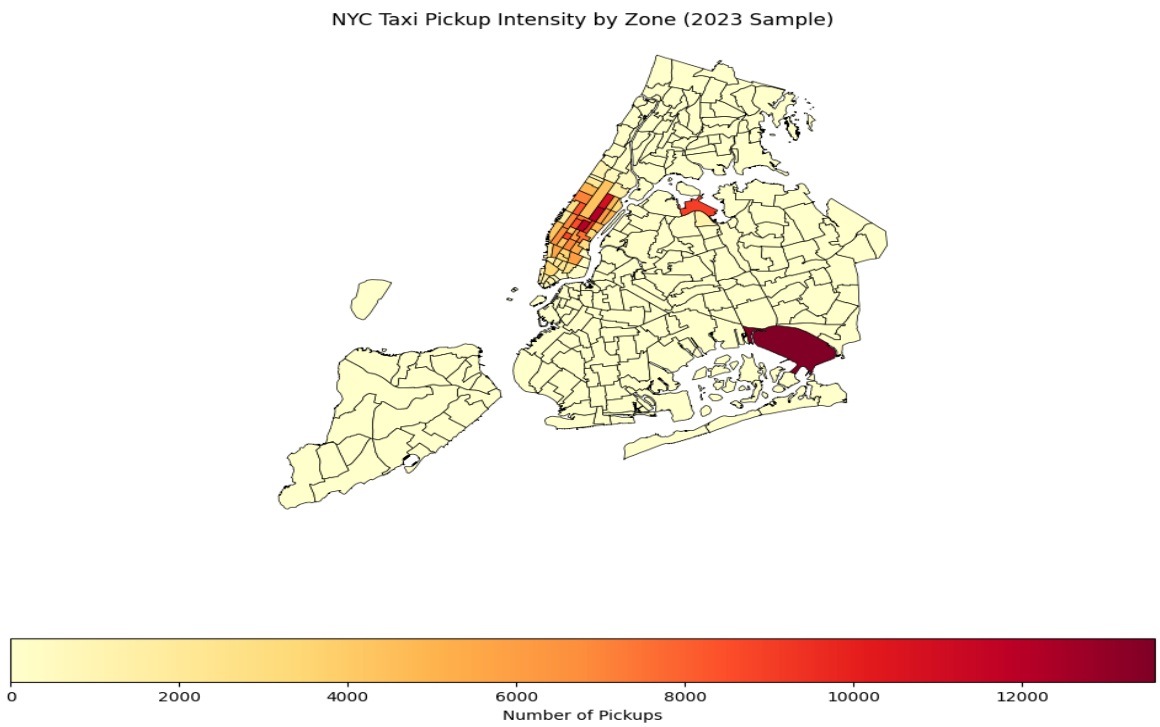

Figure 12: Choropleth map — zones coloured by number of pickups (darker = more trips)

The choropleth map reveals heavy concentration in central Manhattan and the two major airports. Outer-borough zones (Staten Island, eastern Queens, northern Bronx) see minimal yellow taxi activity.

Rank	Zone	Borough	Total Trips (Sample)
1	JFK Airport	Queens	13,580
2	Upper East Side South	Manhattan	12,118
3	Midtown Center	Manhattan	12,032
4	Upper East Side North	Manhattan	10,801
5	Midtown East	Manhattan	9,283
6	LaGuardia Airport	Queens	8,970
7	Penn Station/Madison Sq West	Manhattan	8,746
8	Times Sq/Theatre District	Manhattan	8,566
9	Lincoln Square East	Manhattan	8,357
10	Murray Hill	Manhattan	7,570

Key Insight from General EDA: Taxi operations are overwhelmingly concentrated in Manhattan and the two major airports. These 10 zones alone account for a disproportionate share of revenue. Outer boroughs are largely underserved by yellow taxis relative to their population.

3.2 Detailed EDA: Insights and Strategies

Operational Efficiency

3.2.1 Slow Routes

Average speed (mph) was computed for each route (PULocationID → DOLocationID) per pickup hour as: Speed = trip_distance / (trip_duration / 60). Routes with the lowest average speeds indicate congestion hotspots.

Pickup Hour	Pickup Zone	Dropoff Zone	Avg Speed (mph)
16:00	Central Harlem	Central Harlem	Very Low
0:00	Gramercy	Kips Bay	Low
15:00	West Chelsea/Hudson Yards	Clinton West	Low
20:00	Gramercy	Lower East Side	Low
17:00	Union Sq	Midtown South	Low

Afternoon hours (15:00–17:00) in Midtown Manhattan and neighbouring zones consistently record the lowest speeds. These corridors represent bottlenecks where dispatch time should be factored into estimated arrival calculations.

3.2.2 Hourly Trip Volume

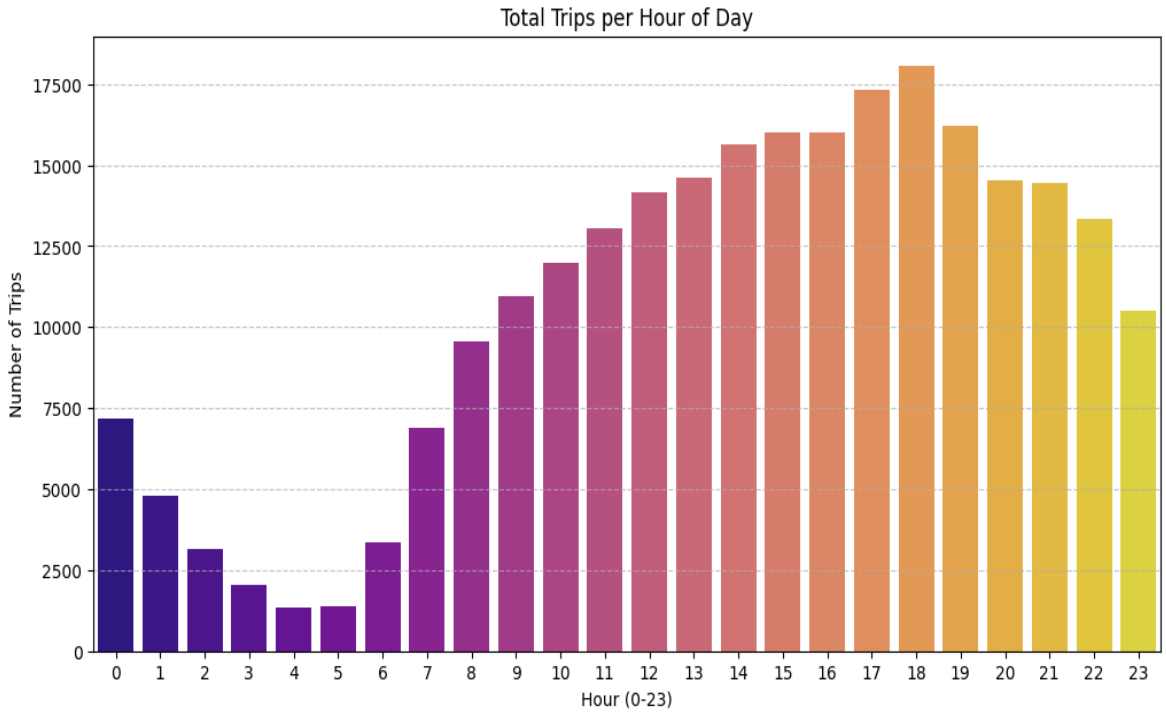

Figure 13: Hourly trip count (sampled) — evening rush at 18:00 is the single busiest hour

The hourly trip distribution confirms the double-peaked pattern of urban taxi demand: a morning rush (08:00–09:00) and a dominant evening rush (17:00–19:00). The 18:00 hour is the single busiest with 18,074 sampled trips.

3.2.3 Scaled Estimates for Busiest Hours

Since the sample represents 0.7% of total 2023 trips, actual volumes are estimated by dividing sampled counts by the sampling fraction (0.007):

Rank	Hour	Sampled Trips	Estimated Actual Trips (2023)
1	18:00	18,074	~2,582,000
2	17:00	17,304	~2,472,000
3	19:00	16,223	~2,318,000
4	15:00	16,024	~2,289,000
5	16:00	15,997	~2,285,000

The 17:00–19:00 window alone accounts for an estimated 7.4 million trips annually — roughly 2% of total NYC yellow taxi trips for the year. This three-hour window deserves maximum fleet deployment.

3.2.4 Weekday vs Weekend Hourly Traffic

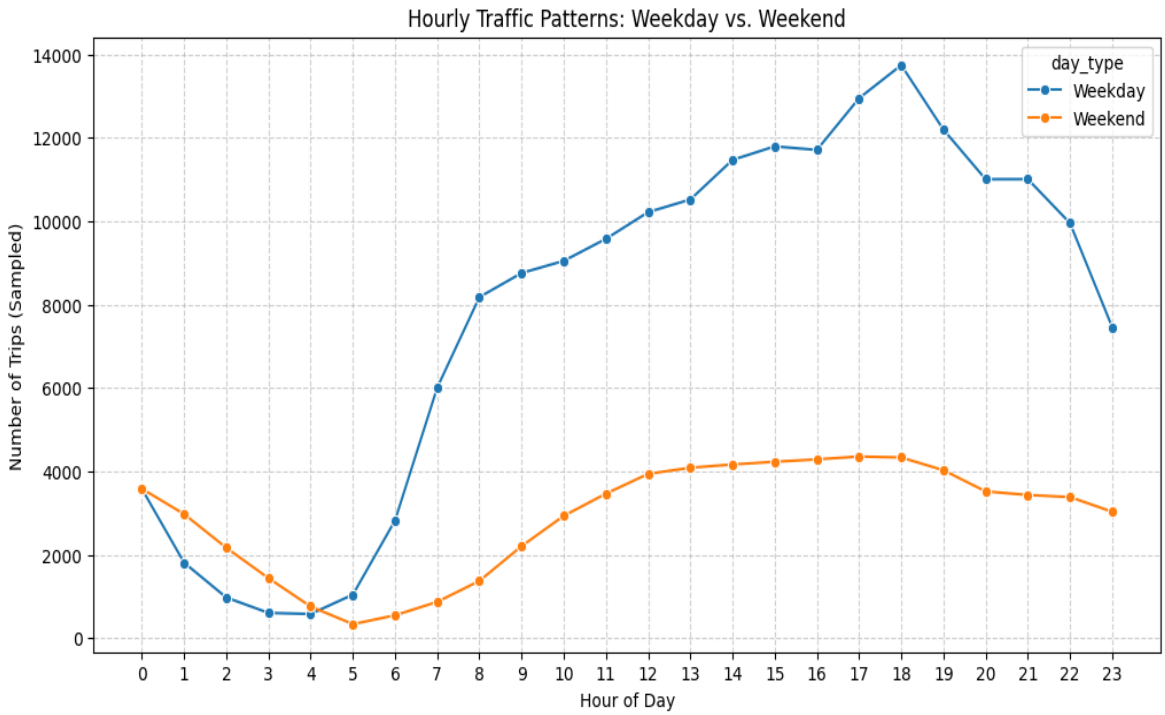

Figure 14: Hourly traffic patterns on weekdays vs weekends — distinct demand profiles

Weekdays show a sharp, pronounced evening peak (17:00–19:00) driven by office commuters. Weekends exhibit a flatter, broader demand curve with elevated late-night activity (00:00–03:00) reflecting nightlife and entertainment trips. Weekend mornings ramp up more slowly — demand does not spike until midday.

3.2.5 Top 10 Pickup and Dropoff Zones

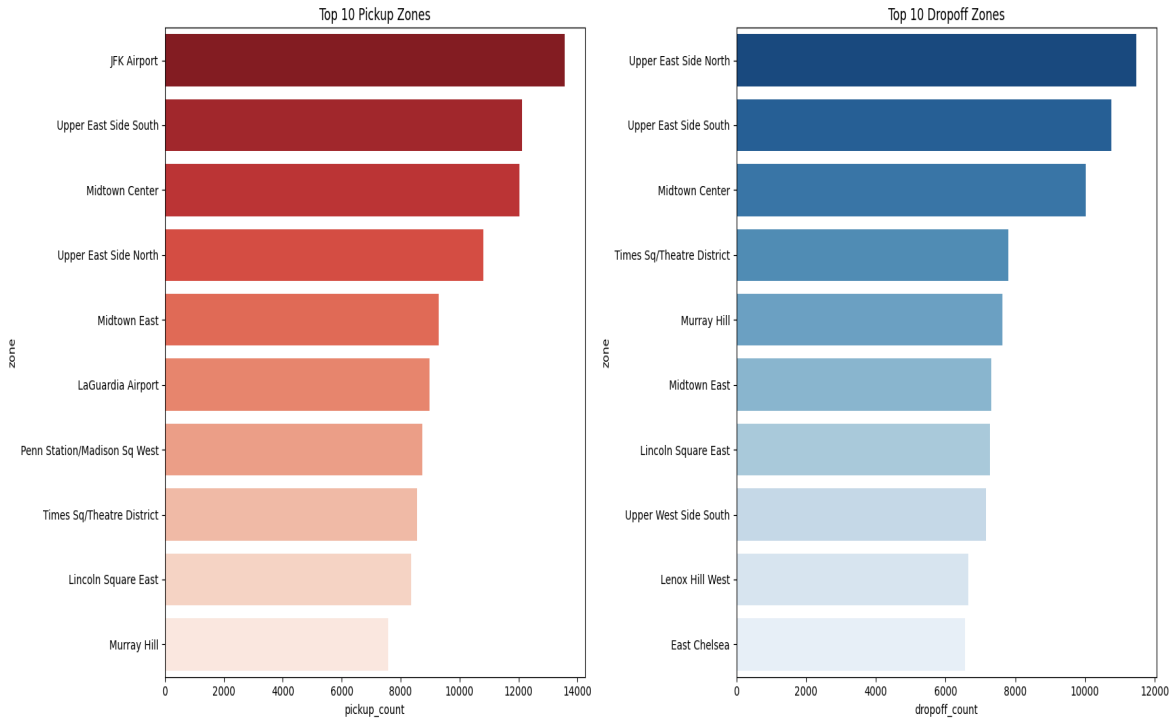

Figure 15: Top 10 zones by hourly pickup count (left) and dropoff count (right)

JFK Airport tops both pickup and dropoff rankings, followed by Upper East Side South and Midtown Center. The same core Manhattan zones dominate both lists, indicating high turnover at these locations. Airport zones (JFK, LaGuardia) stand out as key revenue generators due to longer average trip distances.

3.2.6 Pickup/Dropoff Ratio by Zone

The pickup-to-dropoff ratio reveals directional flow imbalances. A high ratio means more trips originate from a zone than terminate there (net outflow). A low ratio means the zone receives more drop-offs than pick-ups (net inflow).

Zone	P/D Ratio	Interpretation
East Elmhurst	8.09	Major net outflow — airport area generates far more pickups than drop-offs
JFK Airport	4.71	Strong outflow — departing passengers far outnumber arrivals seeking taxis
LaGuardia Airport	2.90	Moderate outflow
Sunset Park East	0.028	Strong net inflow — residents commuting out, few returning by taxi
Windsor Terrace	0.036	Strong net inflow
Whitestone	0.036	Strong net inflow

Airport zones act as major sources of supply (cabs leave full, return empty). Residential outer-borough zones are sinks — cabs drop passengers but rarely pick up there. Strategic repositioning after airport drop-offs to nearby demand centres would reduce deadhead kilometres.

3.2.7 Night-Time Zone Traffic (23:00–05:00)

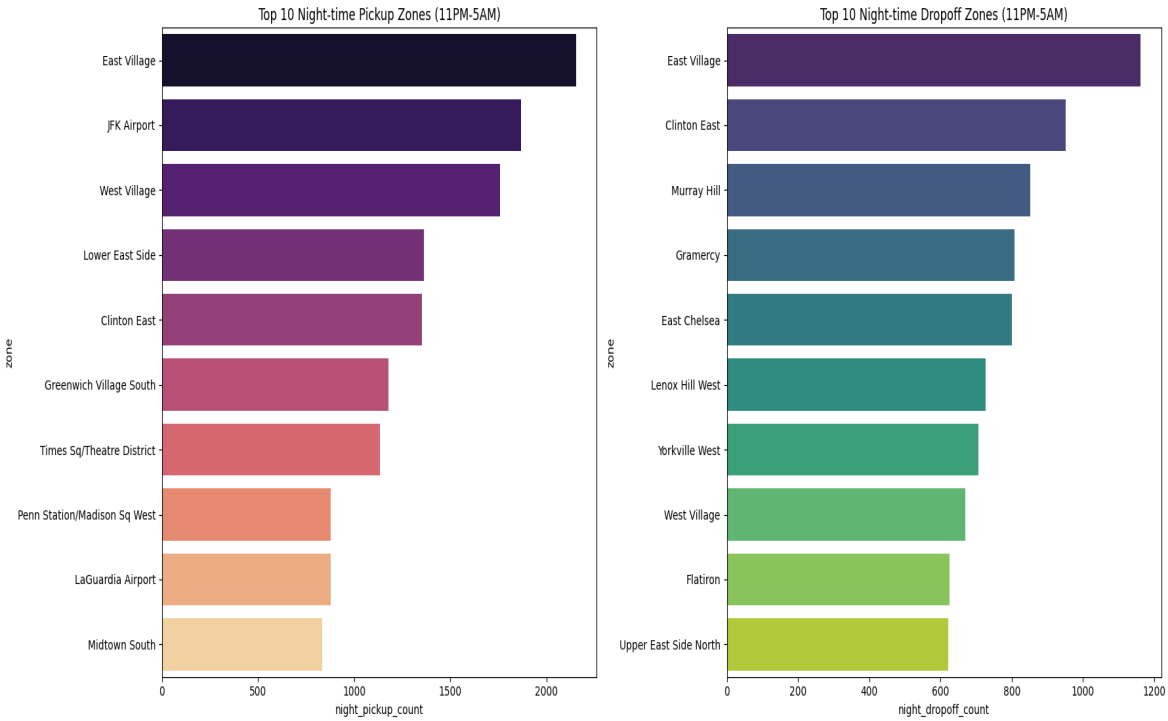

Figure 16: Top 10 zones by night-time pickups and dropoffs (11 PM – 5 AM)

Rank	Zone	Night Pickups
1	East Village	2,154
2	JFK Airport	1,869
3	West Village	1,758
4	Lower East Side	1,363
5	Clinton East	1,354

The East Village, West Village, and Lower East Side are the dominant nightlife clusters driving late-night demand. JFK remains prominent even at night due to red-eye flights. Concentrating night-shift drivers in these zones maximises utilisation.

3.2.8 Revenue — Night vs Day

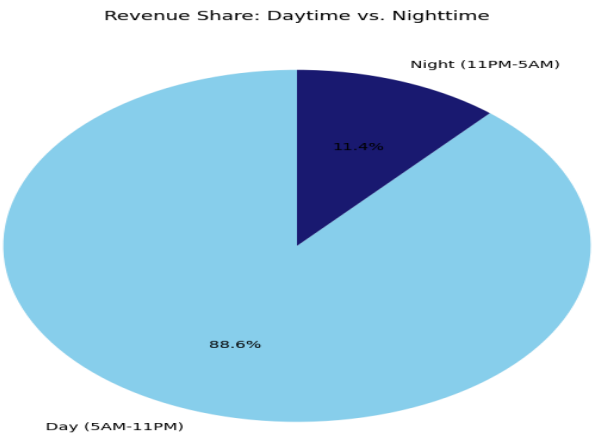

Figure 17: Revenue split between daytime and nighttime hours

Period	Revenue (Sample, $)	Share (%)
Day (05:00–22:59)	$6,557,025	88.6%
Night (23:00–04:59)	$844,256	11.4%

Daytime operations generate 88.6% of revenue. However, the night period's 11.4% share represents significant absolute revenue, and trip length during late-night hours is typically longer (less congestion, more cross-borough journeys), yielding higher per-trip fares.

Pricing Strategy

3.2.9 Fare Per Mile Per Passenger by Passenger Count

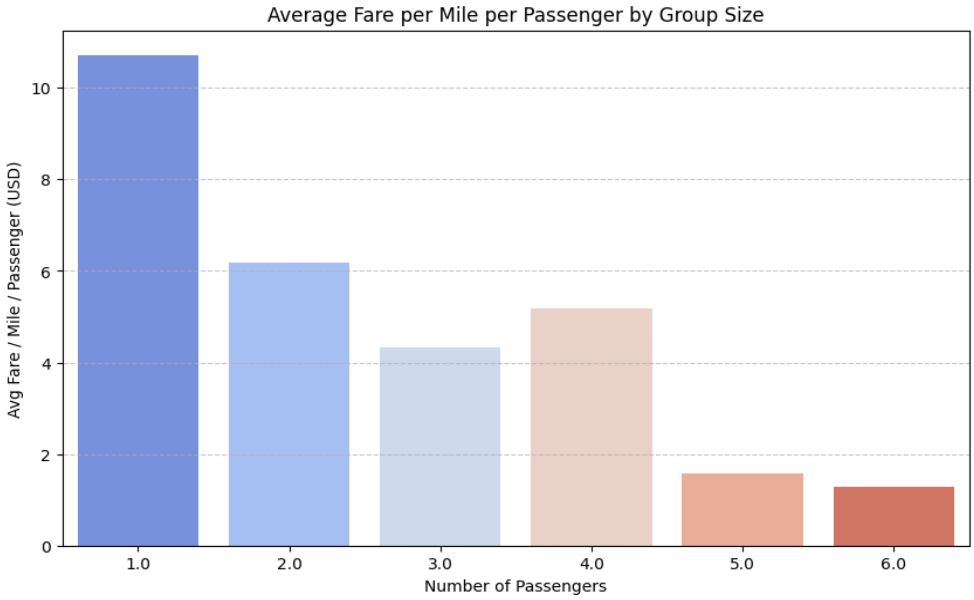

Figure 18: Average fare per mile per passenger — solo rides are significantly more expensive per person

Passenger Count	Avg Fare/Mile/Passenger ($)
1	$10.72
2	$6.19
3	$4.33
4	$5.18
5	$1.57
6	$1.29

Solo riders pay the highest fare per mile per person at $10.72. This drops sharply with additional passengers since the metered fare does not scale with headcount. Group trips offer substantially better value per person — an insight relevant for marketing shared ride options.

3.2.10 Fare Per Mile by Hour and Day

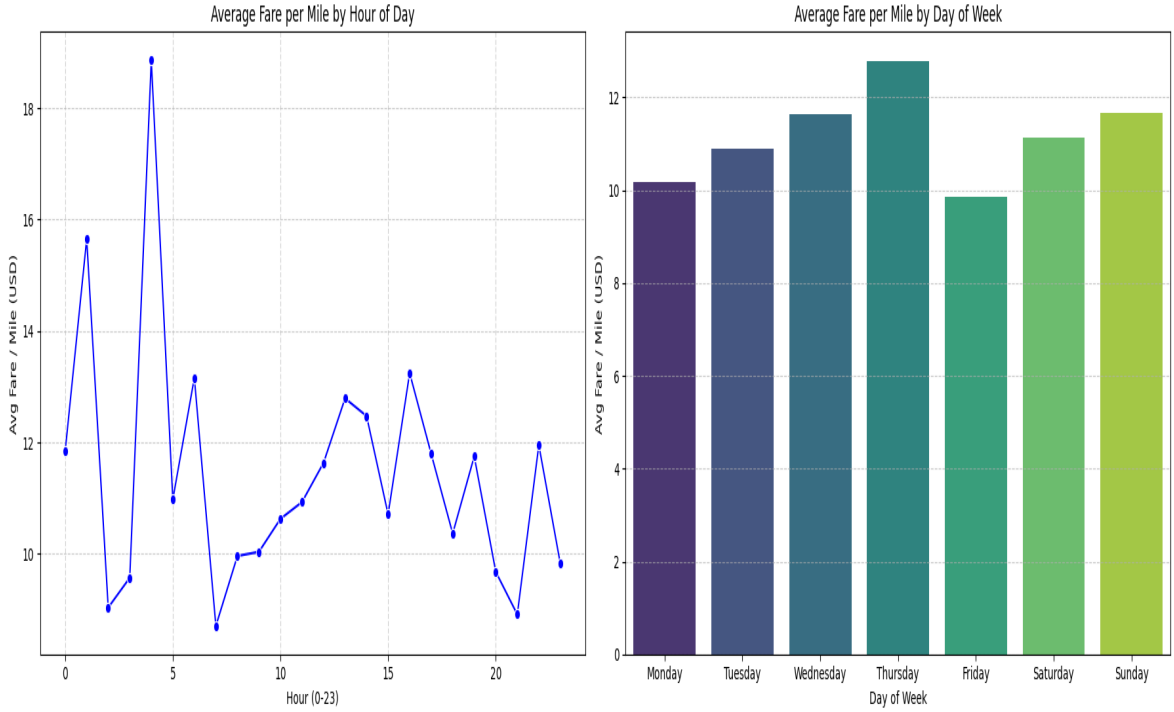

Figure 19: Fare per mile by day of week (left) and top hours (right)

Day	Avg Fare/Mile ($)
Thursday	$12.77 (highest)
Wednesday	$11.65
Sunday	$11.67
Saturday	$11.13
Monday	$10.19 (lowest weekday)
Friday	$9.85 (lowest overall)

Thursday commands the highest average fare per mile, suggesting either more congestion (longer trips in time billing) or longer average trip distances on that day. The early morning hours (01:00–06:00) have the highest fare-per-mile, reflecting longer cross-city journeys in the absence of daytime congestion.

3.2.11 Fare Per Mile by Vendor

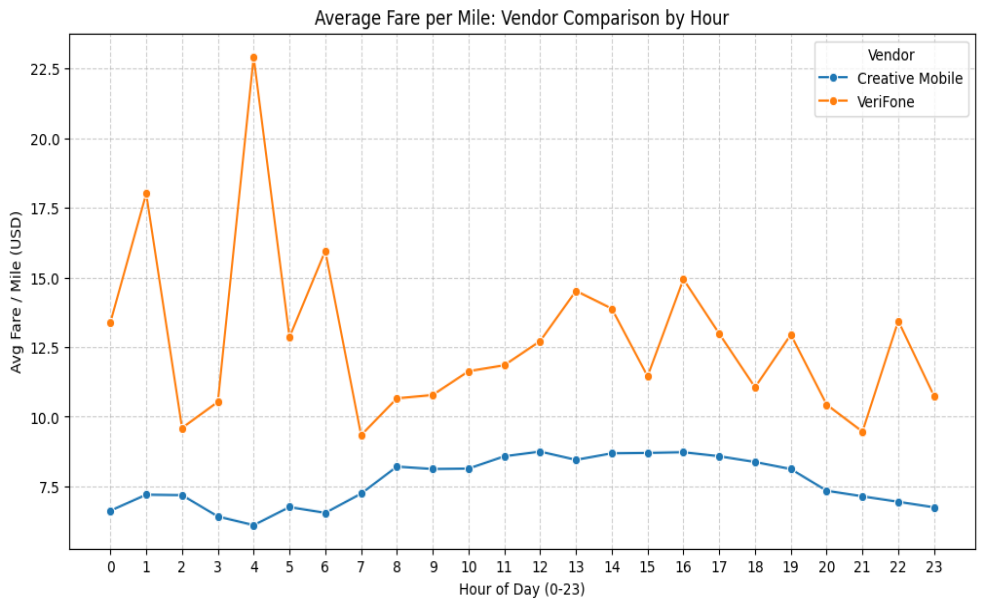

Figure 20: Average fare per mile for Creative Mobile vs VeriFone by hour

Vendor	Avg Fare/Mile ($)
Creative Mobile Technologies (VendorID=1)	$8.03
VeriFone Inc. (VendorID=2)	$12.32

VeriFone-equipped cabs charge 53% more per mile on average than Creative Mobile cabs. This warrants investigation — it may reflect differences in route type (longer vs shorter trips), geographic deployment, or meter calibration differences.

3.2.12 Distance-Tiered Vendor Fare Comparison

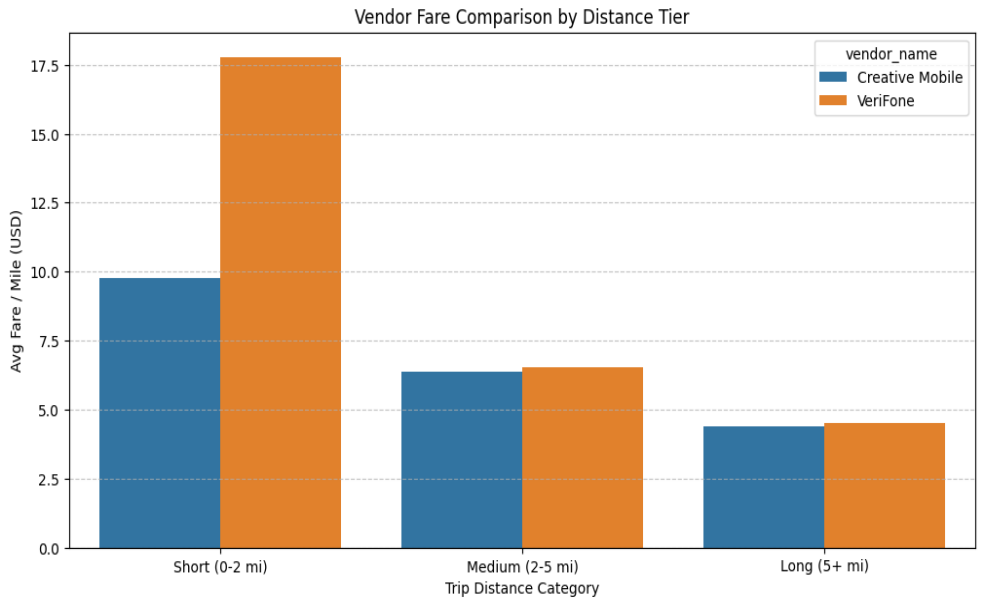

Figure 21: Fare per mile by vendor across distance tiers — VeriFone dominates short trips

Distance Tier	Creative Mobile ($/mile)	VeriFone ($/mile)	VeriFone Premium
Short (0–2 mi)	$9.76	$17.78	+82%

Medium (2–5 mi)	$6.38	$6.54	+2.5%
Long (5+ mi)	$4.41	$4.50	+2.0%

The enormous VeriFone premium on short trips (+82%) is striking. For medium and long trips, the two vendors are nearly identical in pricing. This suggests VeriFone cabs are disproportionately deployed in high-demand, short-distance Manhattan corridors where minimum fare floors or surcharges drive up effective per-mile rates.

Customer Experience

3.2.13 Tip Percentage Analysis

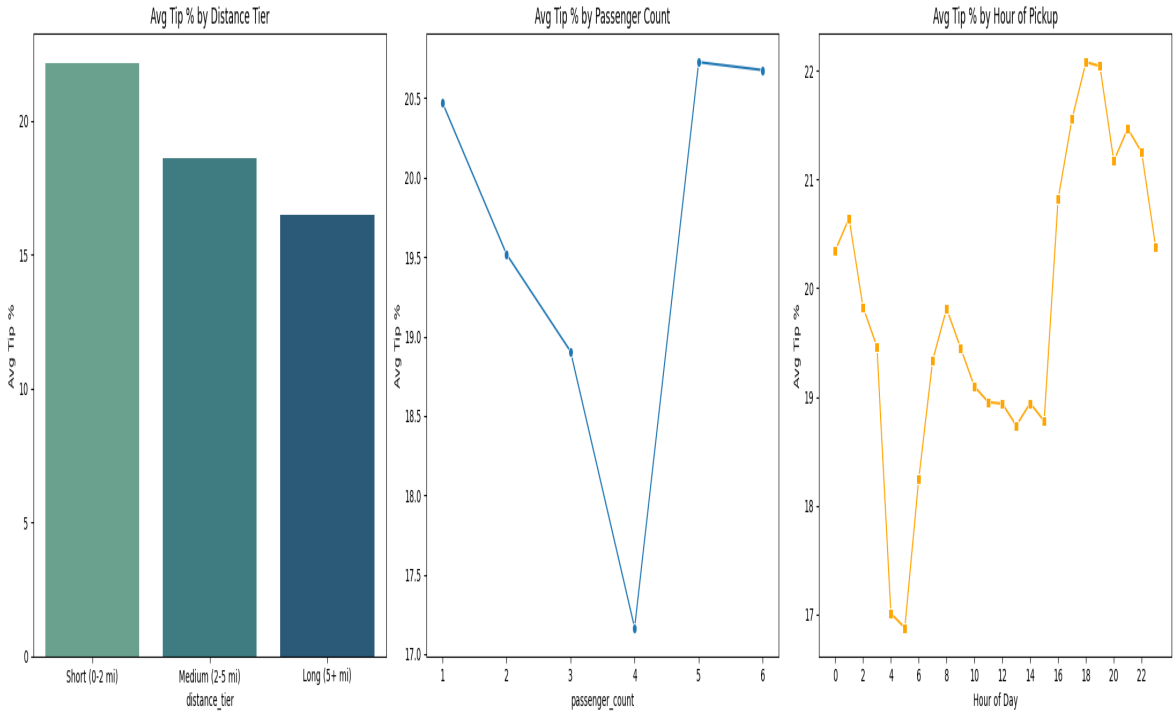

Figure 22: Tip percentage by distance tier, passenger count and hour

Distance Tier	Avg Tip (%)
Short (0–2 mi)	22.1%
Medium (2–5 mi)	18.6%
Long (5+ mi)	16.5%

Short trips attract the highest tip percentages (22%), likely because the absolute tip amount is small regardless (e.g., 22% of $8 = ~$1.76). Passengers on longer trips tend to tip a smaller percentage, though the absolute amount is higher. Drivers benefit from maximising the number of short, high-percentage-tip trips rather than waiting for long-haul fares.

3.2.14 Passenger Count Trends by Hour and Day

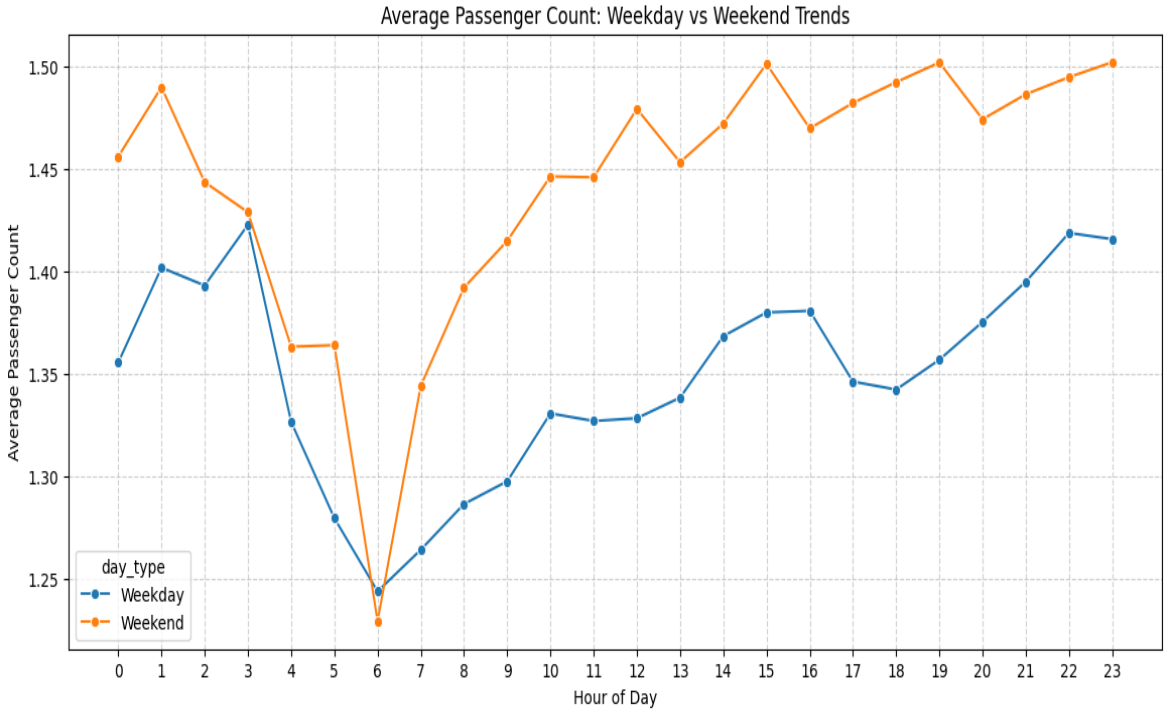

Figure 23: Average passenger count by hour and day type (weekday vs weekend)

Average passenger counts are marginally higher on weekends (Saturday: 1.47, Sunday: 1.46) compared to weekdays (Monday–Thursday: ~1.33–1.36). Friday sees a slight uptick (1.41) consistent with post-work social trips. Evening hours show slightly higher passenger counts regardless of day type.

3.2.15 Passenger Count Variation Across Zones

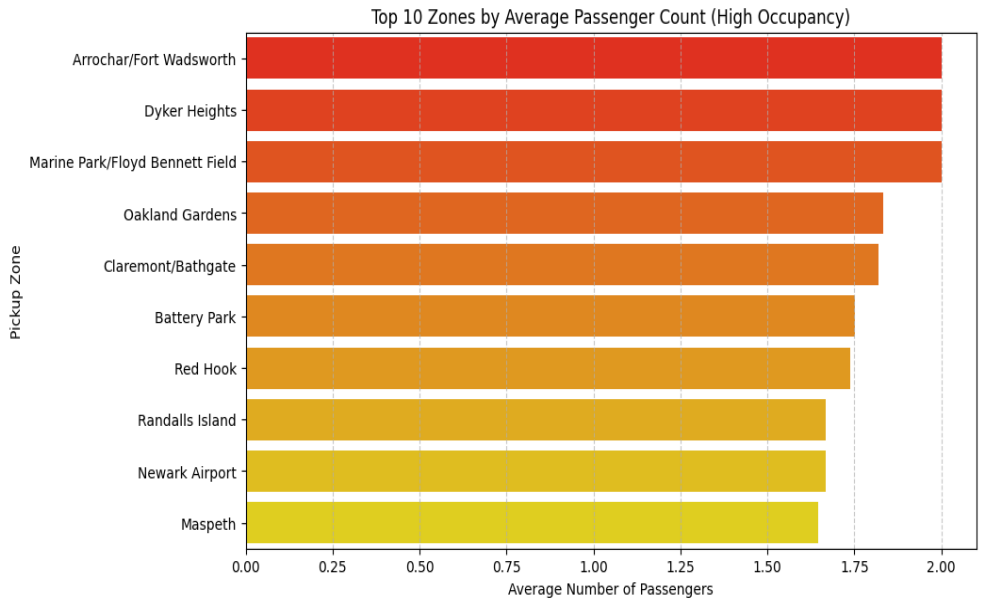

Figure 24: Top zones by average passenger count

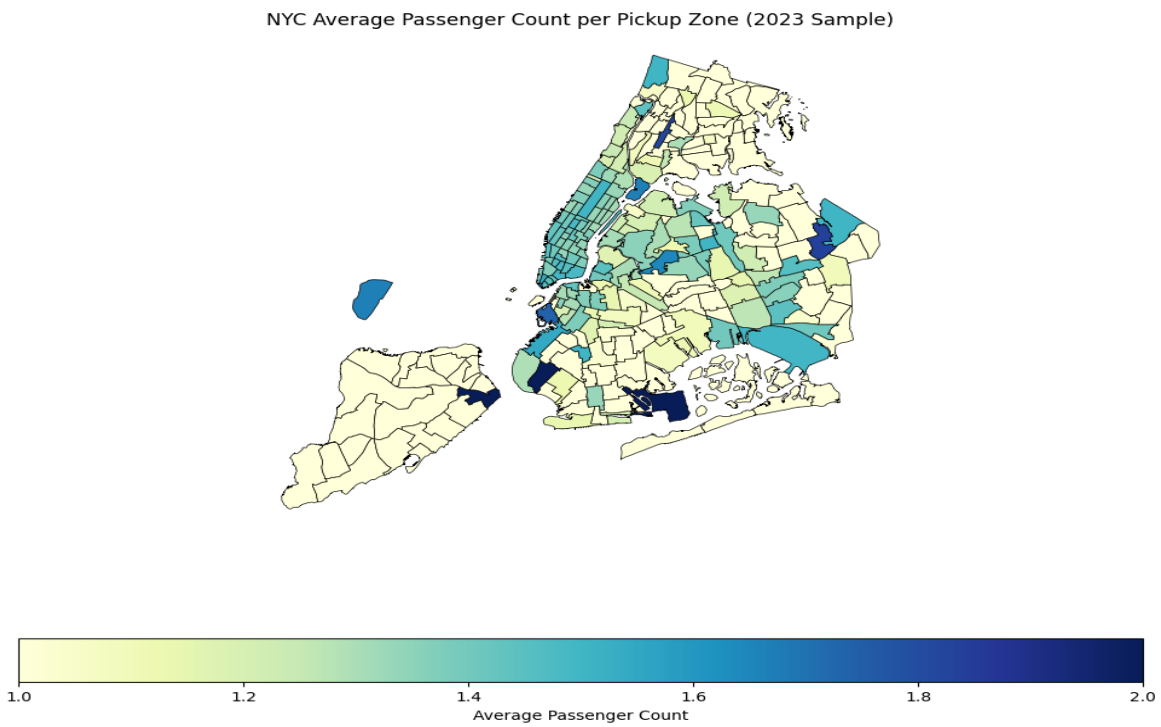

Figure 25: Geographic map of average passenger count by zone

Zone	Avg Passengers
Arrochar/Fort Wadsworth	2.00
Dyker Heights	2.00
Marine Park/Floyd Bennett Field	2.00
Oakland Gardens	1.83
Claremont/Bathgate	1.82
Battery Park	1.75

Outer-borough and suburban-style zones consistently show higher average passenger counts (2.0) — these tend to be residential areas where families or groups share rides to/from transit hubs. Deploying larger vehicles to these zones would improve customer comfort.

3.2.16 Extra Charges and Surcharge Analysis

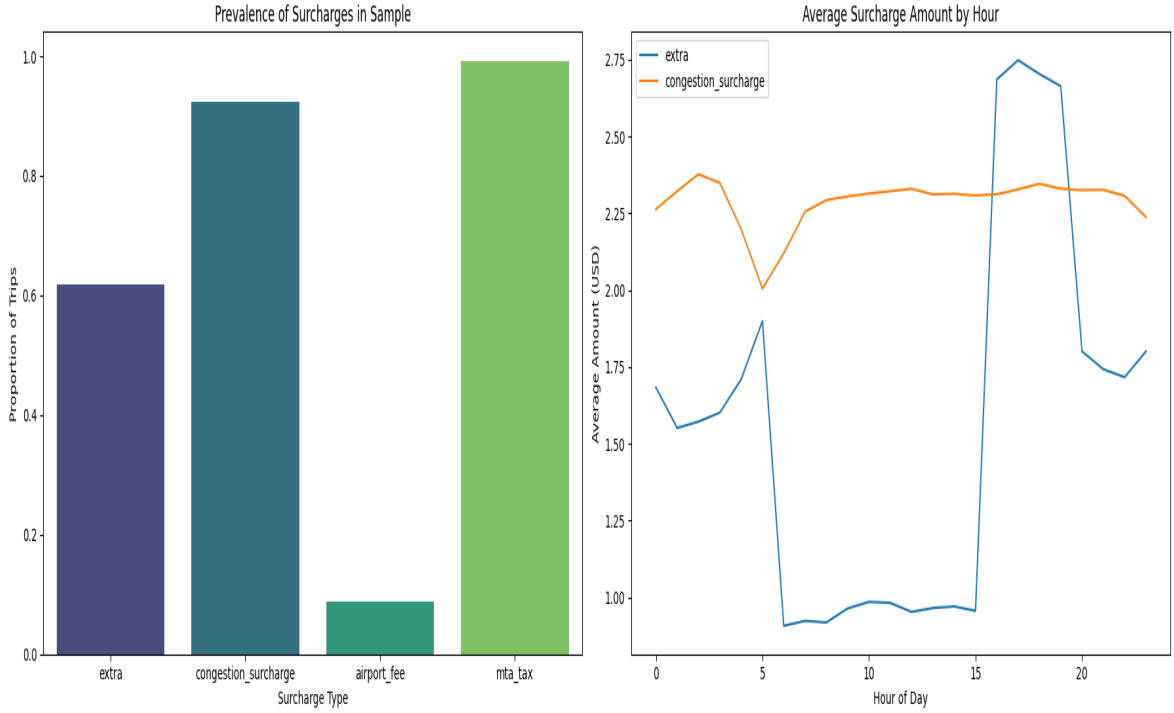

Figure 26: Frequency of extra charges by type and their application patterns

Surcharge Type	Trips Affected	% of Total
MTA Tax ($0.50)	254,193	99.1%
Congestion Surcharge	236,861	92.3%
Extra (Rush hour/night)	158,793	61.9%
Airport Fee ($1.25)	22,595	8.8%

The MTA tax and congestion surcharge are near-universal, applying to almost all trips. The congestion surcharge ($2.50) is applied to trips entering/leaving Manhattan below 96th St. The extra charge (rush hour $1.00, overnight $0.50) applies to 62% of trips. Airport fees are naturally limited to JFK/LaGuardia-adjacent zones (~8.8% of trips).

4. Conclusions and Recommendations

The following strategic recommendations are grounded in the quantitative findings from the exploratory analysis above. Each recommendation maps to specific patterns identified in the data.

4.1 Routing and Dispatching Optimisation

Dispatching decisions should be driven by temporal and spatial demand patterns uncovered in the analysis:

Peak Hour Staging: The 17:00–19:00 window is by far the highest-demand period (estimated 7.4M trips annually across just 3 hours). Drivers should be pre-positioned in Midtown Manhattan, JFK, and LaGuardia Airport from 16:30 onwards to capture this surge without wasted repositioning time.

Avoid Congested Corridors: Mid-town cross-town routes (e.g., Gramercy to Lower East Side at 20:00, West Chelsea at 15:00) record the lowest average speeds (<5 mph). Dispatch algorithms should incorporate real-time speed data for these corridors and route drivers via alternate paths or delay assignment during peak congestion windows.

High-Occupancy Vehicle Deployment: Outer-borough zones (Arrochar, Oakland Gardens, Dyker Heights) consistently see average passenger counts of 2.0. Deploying SUV or minivan vehicles to these zones reduces the incidence of overcrowding complaints and improves passenger satisfaction.

Airport Repositioning: JFK Airport has a pickup-to-dropoff ratio of 4.71 — far more trips originate here than terminate. Drivers completing a drop-off at JFK should be algorithmically directed to the nearest high-demand Midtown or East Village zone rather than returning empty to base.

Night-Shift Focus Zones: East Village, West Village, and Lower East Side are the top night-time pickup zones (23:00–05:00). Dedicating a subset of the night shift fleet to these three zones would capture high-density, high-tip short-haul nightlife trips.

4.2 Strategic Fleet Positioning by Zone

Zone-level analysis reveals distinct demand clusters that should guide where taxis are stationed at different times:

Business Hours (07:00–18:00): Concentrate fleet in the Midtown cluster: Midtown Center, Midtown East, Murray Hill, Lincoln Square East, and Penn Station/Madison Sq West. These five zones collectively handle ~60,000 sampled trips, representing the core of business-hour demand.

Evening Hours (18:00–23:00): Transition focus to Times Square/Theatre District, Penn Station, and Upper East Side South. Theatre shows, restaurant dinner rushes, and commuter overflow sustain high demand through 22:00.

Late Night / Early Morning (23:00–05:00): Redeploy to East Village, West Village, Clinton East, and Lower East Side. These nightlife-heavy zones show the highest late-night pickup concentrations.

Weekend Strategy: Move away from pure business-district focus. Upper East Side North and South, Lincoln Square, and the park-adjacent zones see higher weekend leisure demand. Flexible positioning across these residential/leisure zones on Saturday and Sunday will improve utilisation.

Managing Inflow/Outflow Imbalances: East Elmhurst (ratio 8.09) and JFK Airport (ratio 4.71) are extreme outflow zones. Implementing a driver incentive or platform-managed queuing system at

these airports ensures adequate supply without over-crowding.

4.3 Data-Driven Pricing Strategy

The pricing analysis reveals several actionable opportunities to either capture additional revenue or defend market share against competitors:

Short-Haul Competitive Pricing: VeriFone cabs charge an average of $17.78/mile for short trips (0–2 miles) versus Creative Mobile's $9.76/mile — an 82% premium. New operators should benchmark their short-haul rates closer to $9.76–$10.50/mile to attract price-sensitive urban commuters who compare rates at the kerb.

Peak-Hour Surcharge Retention: The extra charge (rush-hour supplement) currently applies to 61.9% of trips. This surcharge should be maintained during 17:00–19:00 and 00:00–05:00 where supply is constrained and demand is inelastic. Removing it would leave significant revenue on the table.

Driver Earnings Optimisation via Short Trips: Since short-distance trips (0–2 miles) yield the highest tip percentages (22%) and the fastest throughput, encouraging drivers to prioritise high-turnover zones (Midtown, Penn Station) during peak hours will maximise hourly earnings even at a lower per-mile metered rate.

Congestion Surcharge Zone Marketing: 92.3% of trips incur the congestion surcharge. Marketing materials should clearly communicate total trip cost transparency to passengers. Offering loyalty discounts that partially offset the surcharge for frequent riders could improve retention.

Group and Shared Ride Pricing: The fare-per-mile-per-passenger calculation shows group trips (4–6 passengers) cost as little as $1.29–$5.18/person/mile. Formally promoting shared-ride pricing at a slight discount relative to solo fares while maintaining total trip revenue could expand the addressable market in residential zones.

Thursday Premium Opportunity: Thursday consistently shows the highest fare-per-mile ($12.77). This may reflect strong corporate travel or specific route patterns. Analysing Thursday routing in greater depth to understand whether high-value long routes can be replicated on other days would be worthwhile.

4.4 Summary of Key Findings

Finding	Metric / Value
Busiest hour	18:00 (~2.58M estimated annual trips)
Busiest day	Thursday (40,260 sampled trips)
Busiest month	May (23,676 sampled trips)
Peak quarter by revenue	Q2 (27.3% of annual revenue)
Highest-volume zone	JFK Airport (13,580 trips in sample)
Distance–fare correlation	r = 0.943 (very strong)
Payment by credit card	81.5% of all trips
Night revenue share	11.4% of total ($844K in sample)

Top tip rate	22.1% for short trips (0–2 miles)
Vendor pricing gap (short trips)	VeriFone 82% higher than Creative Mobile
Most congested window	15:00–17:00, Midtown/Chelsea corridors

Assumptions declared: (1) 0.7% hourly-stratified sampling is representative of overall 2023 demand; (2) scale-up factor of 1/0.007 ≈ 143× is used for absolute volume estimates; (3) tip amounts from cash trips are not captured in the dataset and are excluded from tip analysis; (4) RatecodeID = 99 is treated as erroneous/missing; (5) trip_duration > 1440 min and trip_distance > 250 miles are treated as data errors.